

NANYANG JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION

Higher 1

MATHEMATICS

8864/01

Paper 1

16 September 2013

3 hours

Additional Materials: Cover sheet
 Answer papers
 Graph paper
 List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your name and class on every script you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagram or graph.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of 6 printed pages.



NANYANG JUNIOR COLLEGE
Internal Examinations

Section A: Pure Mathematics [35 marks]

- 1** Solve the equation

$$\ln(e^{3x} - 3e^{2x}) = x + 2 \ln 2$$

leaving your answer(s) in exact form.

[4]

- 2** Find the range of values of k for which $kx^2 + x + k > 4 - 2x$ for all real values of x .

[4]

- 3** (a) Find the exact value of $\int_0^2 e^{-3x} dx$.

[3]

- (b) Find $\int \left(\frac{a}{(3x+1)} + \frac{x^{5/2}}{\sqrt{x}} \right) dx$, in terms of a .

[2]

- 4** Sketch the graph of $y = 4\sqrt{x} - x$, stating clearly any intercepts with the axes.

[2]

Given that the area bounded by the tangent at the positive x -intercept, the graph $y = 4\sqrt{x} - x$ and the line $x = a$ is given as 1.673 units², find the value of a correct to the nearest integer.

[4]

- 5** The normal to the curve $y = e^{2x} - 7x$ at the point C is parallel to the line $x + y = 0$. Find the exact coordinates of

- (i) the point C ;

[4]

- (ii) the point A where the normal cuts the y -axis.

[2]

Hence find the area of the triangle OAC , where O is the origin.

[2]

- 6 A cylinder with radius r cm and height h cm (see Diagram 1) is placed inside a circular cone of radius 12 cm and height 18 cm so that its base is level with the base of the cone (see Diagram 2).

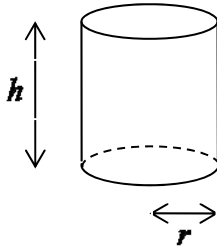


Diagram 1

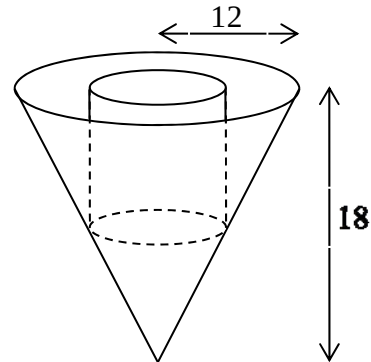


Diagram 2

- (i) By considering a pair of similar triangles, express h in terms of r . [2]
- (ii) Using (i), show that V , the volume of the cylinder, is given by $V = \frac{3\pi r^2(12-r)}{2}$. [1]
- (iii) Find, in exact form, the maximum volume of the cylinder. [5]

Section B: Statistics [60 marks]

- 7 (a) There are 18 people in Millie's class. To choose a person at random, she numbers the people in the class from 1 to 18 and uses the random number generator on her graphing calculator to generate numbers from 1 to 9. Millie then multiplies this number by two and chooses the person corresponding to this new number.

Explain whether this method of choosing a person is random. [2]

- (b) SMRT would like to find out from commuters their views about the new 'Ride for free' scheme. Describe how quota sampling can be used to obtain the views of 100 people for this investigation and describe briefly any advantage that quota sampling has. [4]

- 8** A fair die is thrown three times. Events A and B are defined as follows:

A – a two appears at the first throw,

B – the total score is five.

Calculate $P(B)$ and $P(B|A)$ and hence explain whether A and B are independent events. [5]

- 9** A quest consists of at most 3 stages in a computer game. The probability that a player clears the first stage is $\frac{4}{5}$. For each subsequent stage,

- the conditional probability that a player clears that stage successfully, given that he cleared the preceding stage, is half the probability of success at the preceding stage,
- the conditional probability that a player clears that stage, given that he failed at the preceding stage, is $\frac{1}{3}$.

The quest ends if a player fails to clear 2 consecutive stages, and a player is deemed to have successfully cleared the quest if he or she clears 2 stages.

(i) Draw a probability tree diagram to represent the information above. [2]

(ii) Find the probability that a player clears the quest successfully. [3]

A total of 3 players attempt the quest. Find the probability that at least 2 of the three players successfully clear the quest. [3]

- 10** A factory produces LCD projectors, some of which are defective. On average, it is found that 8% of the projectors are defective. A quality controller examined a random sample of 20 projectors.

(i) Calculate the probability that the number of defective projectors is not more than 2. [2]

(ii) Given that there are more than 3 defective projectors in a sample of 20 projectors, find the probability that there are at most 6 defective projectors. [3]

(iii) A distributor bought 65 batches of 20 projectors each. Find the probability that the mean number of defective projectors in each batch is not more than 2. [3]

The quality controller decided to take another random sample of 120 projectors. Using a suitable approximation, find the largest value of a such that the probability that the number of defective projectors is at least a is more than 0.6. [5]

- 11** The random variable X is normally distributed with mean μ and standard deviation σ . It is given that $3\mu = 7\sigma^2$ and that $P(X > 2\mu) = 0.1016$. Find μ and σ . [4]
- 12** Cans of drinks are packed in boxes of 4 cans. The mass of a randomly chosen can of drink is normally distributed with mean 510 g and standard deviation 14 g. The mass of a randomly chosen box, when empty, is normally distributed with mean 200 g and standard deviation 8 g.
- (i) Find the probability that the total mass of a full box of cans of drinks is between 2200 g and 2300 g. [4]
 - (ii) State an assumption needed for your calculations in (i). [1]
 - (iii) 2 cans of drinks are chosen at random. Find the probability that they differ in mass by more than 20 g. [4]
- 13** An experiment was undertaken to collect information on the burning of a specific type of wood as a source of energy. At given fixed levels of the wood's moisture content, x percent, its corresponding calorific value, y MWh/tonne, on burning was determined. The results are shown in the table.
- | | | | | | | | | | | | | | |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| x | 5 | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50 | 55 | 60 | 65 |
| y | 5.2 | 4.7 | 4.3 | 4.0 | 3.2 | 2.8 | 2.5 | 2.2 | 1.8 | 1.5 | 1.3 | 1.0 | 0.6 |
- (i) Explain why x is the independent variable. [1]
 - (ii) Find the equation of the regression line of y on x in the form $y = a + bx$. [1]
 - (iii) Interpret the values for a and b in the context of the question. [2]
 - (iv) Use your equation to estimate the wood's calorific value when it has a moisture content of 27 percent. [1]
 - (v) Based on the context of this question and with numerical support, give a reason why your equation cannot be used to estimate the wood's calorific value when it has a moisture content of 80 percent. [2]

- 14** A company has machines which produce and dispense beverages at an average of 330 ml per can. Not long after changing their machines to new ones, there are complaints that the can drinks contain less than 330ml of beverage. The company decided to take a sample of 80 cans to investigate and the data are summarized below, where x is the amount of beverage in a can.

$$\sum x = 26080 \qquad \sum x^2 = 8530200$$

- (i)** Find the unbiased estimates of the population mean and variance of the amount of beverage in the cans. **[2]**
- (ii)** Test at 5% significance level whether the new machines dispense less beverages than the old machines. **[5]**
- (iii)** Explain if it is necessary to assume that the amount of beverage dispensed by the machine follows a normal distribution in making the calculation in **(ii)**. **[1]**

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